

IEEE 754 Double-Precision Floating-Point Multiplier

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I. ARCHITECTURE

The IEEE 754 double-precision floating-point multiplier operates on 64-bit operands conforming to the standard format: 1 sign bit, 11 exponent bits (biased by 1023), and 52 explicit mantissa bits with an implicit leading 1 for normalized numbers. The top-level architecture is illustrated in Fig. 1.

Inputs A and B enter the datapath in parallel. The output sign bit Z[63] is computed combinatorially as the XOR of A[63] and B[63]. The 11-bit exponents are summed by a dedicated carry-lookahead adder, producing a 13-bit intermediate sum to accommodate overflow detection. The 53-bit mantissas are forwarded to the Wallace tree multiplier, producing a 106-bit product.

Both intermediate results are passed to the Normalizer, which handles special-case detection (zero, infinity, NaN), exponent normalization, denormal handling, and IEEE 754 round-to-nearest-even. The final 64-bit result Z is produced at the output of the Normalizer.

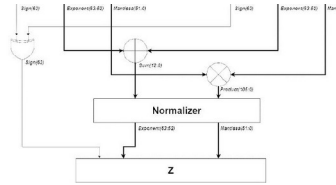


Fig. 1. Top-level architecture of the double-precision FP multiplier.

II. MANTISSA MULTIPLIER

Multiplying two 53-bit mantissas requires a large partial product array. This design employs a Wallace tree reduction architecture to minimize adder depth, achieving $O(\log n)$ carry-save reduction levels rather than the $O(n)$ depth of a ripple-carry array multiplier.

The two 53-bit operands—{1'b1, A_MANTISSA[51:0]} and {1'b1, B_MANTISSA[51:0]}—generate 53 partial products via AND gates. The Wallace tree uses a cascade of Carry-Save Adders (CSAs) to reduce the partial product array. Each CSA accepts three rows and produces two (a sum and a carry), reducing the count by one-third per stage.

The reduction tree requires 10 pipeline stages to collapse all 53 partial products down to a final Sum and Carry vector, each 106 bits wide. A Ripple-Carry Adder (RCA) combines these to produce the 106-bit product

III. NORMALIZATION

The Normalizer post-processes the raw exponent sum and mantissa product to produce a correctly formatted IEEE 754 result. It handles four cases: special inputs, exponent overflow, exponent underflow (denormal), and the normal case.

After the exponent adder computes Sum[12:0] and the mantissa multiplier produces Product[105:0], the Normalizer locates the leading '1' in Product. This position determines the shift amount Shift[9:0] applied to align the mantissa to the normalized form 1.xxx. The normalized exponent NORMALIZED_EXPONENT[12:0] is obtained by adding the shift amount to the raw exponent sum, accounting for the IEEE 754 exponent bias.

The Normalized Exponent is checked for overflow (value > 2046 , representing $\pm\infty$) and underflow (value ≤ 0 , representing a denormal or zero). In the overflow case, the output is forced to the appropriate infinity. In the underflow case, the mantissa is right-shifted until the biased exponent reaches zero, producing a denormal result, followed by rounding. In the normal case, the 52-bit mantissa is rounded using round-to-nearest-even and output directly. The complete datapath is shown in Fig. 3.

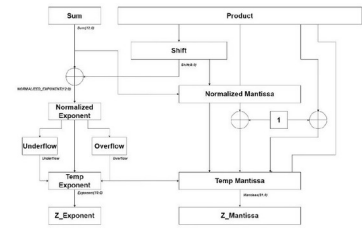


Fig. 3. Normalization datapath showing exponent and mantissa pipelines.

The control flow is shown in Fig. 4. Special input detection occurs first; if either input is NaN, infinity, or zero, the appropriate result is produced without invoking the multiplier result. Otherwise the exponent normalization path is taken. Rounding in both the normal and denormal paths follows IEEE 754 round-to-nearest-even: if the guard and round bits are 1 and the sticky bit is 1, or if rounding is exactly halfway and the LSB of the result is 1, the mantissa is incremented. This may overflow the 52-bit mantissa field, in which case the exponent is also incremented.

Product[105:0]. Pipeline registers between stages ensure the critical path meets the 2.35 GHz timing closure target.

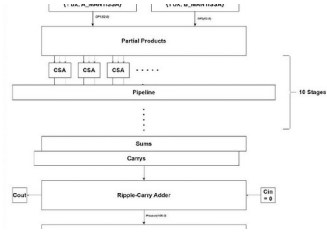


Fig. 2. Wallace tree multiplier with 10-stage CSA pipeline.

IV. IMPLEMENTATION RESULTS

A. Synthesis Summary

The design was synthesized using Genus Synthesis Solution 20.10 targeting the TSMC 16nm N16ADFP standard cell library at the ss0p72v125c corner. Key results are summarized in Table I.

Parameter	Value
Process	TSMC 16 nm
Max Clock Freq.	2.35 GHz
Cell Count	18,683
Cell Area	14,238.478 μm^2
Net Area	3,121.197 μm^2
Total Area	17,359.675 μm^2
Dynamic Power	64.8 mW
Leakage Power	5 mW
Timing Slack	+41 ps

Table I. Synthesis summary.

Timing closure is achieved with +41 ps of positive slack at the critical endpoint CORE_LOC_reg[2]/D (clock group CLK). Both max-transition and max-capacitance design rules pass without violations, confirming the design is DRC clean.

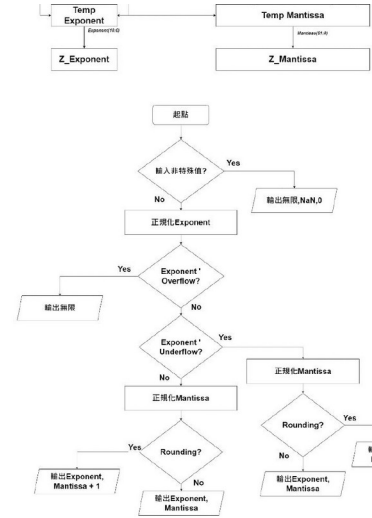


Fig. 4. Normalization control flow.

B. Power Analysis

Power was analyzed using PDB frame-based simulation. Register logic dominates at 88.92% of total power, consistent with the deeply pipelined 10-stage Wallace tree. Combinational logic accounts for the remaining 11.00%.

Category	Total (W)	Row %
register	6.22e-02	88.92%
logic	7.75e-03	11.00%
memory	0.00e+00	0.00%
Subtotal	6.99e-02	100%

Table II. Power breakdown by cell category.

C. IR Drop Distribution

IR drop analysis shows 38.75% of cells fall within the 0.00–2.44 mV range, and 10.20% show negative IR drop indicating local power supply boosting. Only 3.55% exceed 17.07 mV, with a single cell above 19.51 mV, demonstrating a well-distributed power delivery network.

Range (mV)	Count	%
> 19.51	1	0.00%
17.07–19.51	7,909	3.55%
14.63–17.07	17,893	8.04%
12.19–14.63	17,457	7.85%
9.75–12.19	17,273	7.76%
7.32–9.75	16,791	7.55%
4.88–7.32	16,035	7.21%

2.44–4.88	20,221	9.09%
0.00–2.44	86,209	38.75%
< 0.00	22,692	10.20%

Table III. IR drop distribution.